AD23532 – PRINCIPLES OF DATA SCIENCE

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EXPERIMENT 1 – DATA PREPROCESSING AND CLEANING

Problem Statement

A telecommunications company faces customer churn. As a Data Scientist, preprocess customer data by handling missing values, removing inconsistencies, encoding categorical features, scaling numerical attributes, and visualizing important relationships before building a churn prediction model.

Learning Objective

To understand and perform essential data preprocessing techniques to prepare high-quality data for machine learning and predictive analytics.

Tasks to Perform

1. Import the required Python libraries.
2. Load the Telco Customer Churn Dataset into a Pandas DataFrame. And display First 10 records, Last 10 records, Shape of the dataset, Column names.
3. Explore the Dataset and display Dataset information, Summary statistics, Number of missing values, Number of duplicate records
4. Clean the data
   a. Remove duplicate customer records.
   b. Replace missing values i. Numerical columns → Median ii. Categorical columns → Mode.
   c. Verify that no missing values remain.
5. Encode categorical columns: Gender, Partner, Dependents, Contract, PaymentMethod, InternetService, Churn. Use appropriate encoding techniques.
6. Standardise the numerical features: Tenure, MonthlyCharges, TotalCharges, using StandardScaler.
7. Plot Distribution of MonthlyCharges, Distribution of Tenure.
8. Create a Count Plot showing Contract vs Churn.
9. Draw a Box Plot of MonthlyCharges grouped by Churn.
10. Display the Correlation Heatmap of all numerical features.

AIM

To preprocess the Telco Customer Churn dataset by handling missing values, removing duplicates, encoding categorical features, scaling numerical features, and visualizing data relationships.

ALGORITHM

1. Import the required Python libraries and load the dataset.
2. Display the first 10 records, dataset information, and summary statistics.
3. Remove duplicate records and handle missing values using median and mode.
4. Encode categorical features and scale numerical features using `StandardScaler`.
5. Plot distributions of MonthlyCharges and Tenure.
6. Create a count plot of Contract vs Churn and a box plot of MonthlyCharges grouped by Churn.
7. Display the correlation heatmap of numerical features.

PROGRAM AND OUTPUT


```
# Question 1: Import the required Python libraries and load the Telco Customer Churn dataset into a Pandas DataFrame.
# LUQMAN T (241801141)
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler, LabelEncoder

# Update the path/URL if your dataset is stored elsewhere.
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
print(df.head())
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes

               PaymentMethod MonthlyCharges  TotalCharges Churn
0           Electronic check          29.85         29.85    No
1               Mailed check          56.95        1889.5    No
2               Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4           Electronic check          70.70        151.65   Yes

[5 rows x 21 columns]
```

```
# Question 2: Display the first 10 records, last 10 records, dataset shape, and column names.
# LUQMAN T (241801141)
print("First 10 records:")
display(df.head(10))

print("Last 10 records:")
display(df.tail(10))

print("Dataset shape:", df.shape)
print("Column names:", df.columns.tolist())
```

First 10 records:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecur
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	
5	9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	
6	1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	
7	6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	
8	7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	
9	6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	

10 rows × 21 columns

Last 10 records:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSe
7033	9767-FFLEM	Male	0	No	No	38	Yes	No	Fiber optic	
7034	0639-TSIQW	Female	0	No	No	67	Yes	Yes	Fiber optic	
7035	8456-QDAVC	Male	0	No	No	19	Yes	No	Fiber optic	
7036	7750-EYXWZ	Female	0	No	No	12	No	No phone service	DSL	
7037	2569-WGERO	Female	0	No	No	72	Yes	No	No	No
7038	6840-RESVB	Male	0	Yes	Yes	24	Yes	Yes	DSL	
7039	2234-XADUH	Female	0	Yes	Yes	72	Yes	Yes	Fiber optic	
7040	4801-JZAZL	Female	0	Yes	Yes	11	No	No phone service	DSL	
7041	8361-LTMKD	Male	1	Yes	No	4	Yes	Yes	Fiber optic	
7042	3186-AJIEK	Male	0	No	No	66	Yes	No	Fiber optic	

10 rows × 21 columns

```
Dataset shape: (7043, 21)
Column names: ['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines',
```

```
# Question 3: Explore the dataset using summary statistics and identify missing values and duplicate records.
# LUQMAN T (241801141)
print("Summary statistics:")
display(df.describe(include="all"))

print("Missing values per column:")
print(df.isnull().sum())

print("Number of duplicate records:", df.duplicated().sum())
```

Summary statistics:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	Or
count	7043	7043	7043.000000	7043	7043	7043.000000	7043	7043	7043	
unique	7043	2	NaN	2	2	NaN	2	3	3	
top	3186-AJIEK	Male	NaN	No	No	NaN	Yes	No	Fiber optic	
freq	1	3555	NaN	3641	4933	NaN	6361	3390	3096	
mean	NaN	NaN	0.162147	NaN	NaN	32.371149	NaN	NaN	NaN	
std	NaN	NaN	0.368612	NaN	NaN	24.559481	NaN	NaN	NaN	
min	NaN	NaN	0.000000	NaN	NaN	0.000000	NaN	NaN	NaN	
25%	NaN	NaN	0.000000	NaN	NaN	9.000000	NaN	NaN	NaN	
50%	NaN	NaN	0.000000	NaN	NaN	29.000000	NaN	NaN	NaN	
75%	NaN	NaN	0.000000	NaN	NaN	55.000000	NaN	NaN	NaN	
max	NaN	NaN	1.000000	NaN	NaN	72.000000	NaN	NaN	NaN	

11 rows × 21 columns

```
Missing values per column:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
Number of duplicate records: 0
```

```
# Question 4: Remove duplicate records and replace missing numerical values with the median and categorical values with th
# LUQMAN T (241801141)
df = df.drop_duplicates().copy()

for col in df.columns:
    if df[col].isnull().any():
        if pd.api.types.is_numeric_dtype(df[col]):
            df[col] = df[col].fillna(df[col].median())
        else:
            mode_values = df[col].mode(dropna=True)
            if not mode_values.empty:
                df[col] = df[col].fillna(mode_values.iloc[0])

print("Duplicates removed. Dataset shape:", df.shape)
```

```
Duplicates removed. Dataset shape: (7043, 21)
```

```
# Question 5: Verify that no missing values remain in the dataset.
# LUQMAN T (241801141)
print("Missing values per column:")
print(df.isnull().sum())
print("Total missing values:", df.isnull().sum().sum())
```

```
Missing values per column:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
```

```
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
Total missing values: 0
```

```
# Question 6: Encode categorical columns using appropriate encoding techniques.
# LUQMAN T (241801141)
categorical_cols = df.select_dtypes(include=["object", "category"]).columns.tolist()

# One-hot encode nominal columns; this also encodes the target column Churn.
df_encoded = pd.get_dummies(df, columns=categorical_cols, drop_first=True, dtype=int)

print("Encoded dataset shape:", df_encoded.shape)
display(df_encoded.head())
```

```
Encoded dataset shape: (7043, 13602)
```

	SeniorCitizen	tenure	MonthlyCharges	customerID_0003-MKNFE	customerID_0004-TLHLJ	customerID_0011-IGKFF	customerID_0013-EXCHZ	customerID
0	0	1	29.85	0	0	0	0	
1	0	34	56.95	0	0	0	0	
2	0	2	53.85	0	0	0	0	
3	0	45	42.30	0	0	0	0	
4	0	2	70.70	0	0	0	0	

5 rows × 13602 columns

```
# Question 7: Standardize Tenure, MonthlyCharges, and TotalCharges using StandardScaler.
# LUQMAN T (241801141)
# Convert TotalCharges to numeric if it contains blank strings.
for col in ["tenure", "MonthlyCharges", "TotalCharges"]:
    if col in df.columns:
        df[col] = pd.to_numeric(df[col], errors="coerce")
        if df[col].isnull().any():
            df[col] = df[col].fillna(df[col].median())

numeric_features = ["tenure", "MonthlyCharges", "TotalCharges"]
scaler = StandardScaler()
df[numeric_features] = scaler.fit_transform(df[numeric_features])

print("Standardized numerical features:")
display(df[numeric_features].head())
```

```
Standardized numerical features:
```

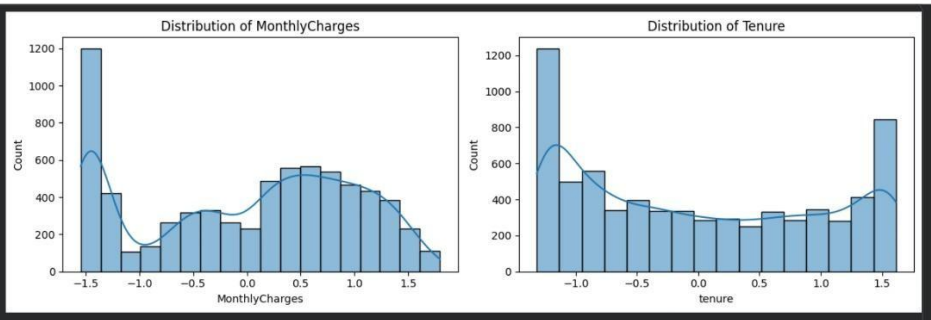
	tenure	MonthlyCharges	TotalCharges
0	-1.277445	-1.160323	-0.994242
1	0.066327	-0.259629	-0.173244
2	-1.236724	-0.362660	-0.959674
3	0.514251	-0.746535	-0.194766
4	-1.236724	0.197365	-0.940470

```
# Question 8: Plot the distributions of MonthlyCharges and Tenure using histograms.
# LUQMAN T (241801141)
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

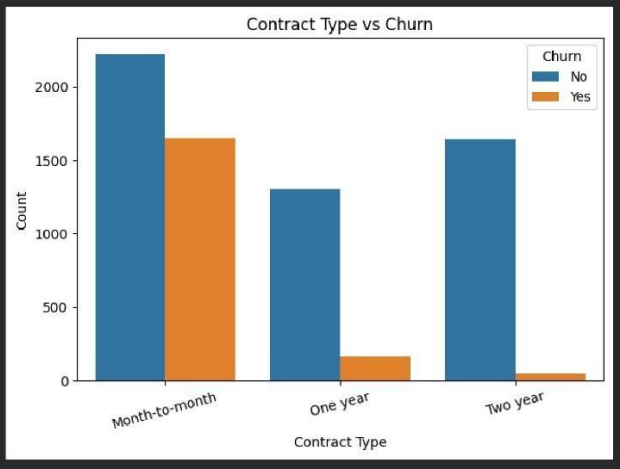
sns.histplot(df["MonthlyCharges"], kde=True, ax=axes[0])
axes[0].set_title("Distribution of MonthlyCharges")

sns.histplot(df["tenure"], kde=True, ax=axes[1])
axes[1].set_title("Distribution of Tenure")

plt.tight_layout()
plt.show()
```




```
# Question 9: Create a count plot showing Contract type against Churn.
# LUQMAN T (241801141)
sns.countplot(data=df, x="Contract", hue="Churn")
plt.title("Contract Type vs Churn")
plt.xlabel("Contract Type")
plt.ylabel("Count")
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()
```

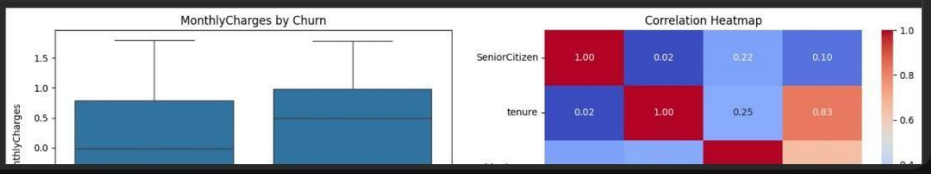



```
# Question 10: Draw a box plot of MonthlyCharges grouped by Churn and generate a correlation heatmap of all numerical featu
# LUQMAN T (241801141)
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.boxplot(data=df, x="Churn", y="MonthlyCharges", ax=axes[0])
axes[0].set_title("MonthlyCharges by Churn")

numeric_df = df.select_dtypes(include=np.number)
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm", fmt=".2f", ax=axes[1])
axes[1].set_title("Correlation Heatmap")

plt.tight_layout()
plt.show()
```



RESULT

Thus, the Telco Customer Churn dataset has handled missing values, removed duplicates and performed all the tasks and executed successfully.